
THE ONTOLOGY OF THE ALIEN: WORLD-DIVERSITY SEARCH AND EVOLVING SOLUTION ONTOLOGIES

A PREPRINT

Henrik Westerberg
Emergent Wisdom
henrik.westerberg@emergentwisdom.org

March 7, 2026
Revised: August 15, 2026

ABSTRACT

Open-ended problems such as institutional design lack enumerable solution spaces and cheap automatic evaluators. *World-diversity search* moves variation upstream. Target-independent Builders construct textual worlds with altered causal rules; purpose-blind Solvers address the same target under each rule set; and compilers translate the resulting mechanisms into a common target-domain schema. Building on this previously released operator, this paper introduces *ontology-governed intervention search*. A Taxonomist curates a persistent typed ontology of complete interventions, judges structural equivalence, and can turn rejection into targeted commissioning: its diagnosis specifies which causal relation the next proposal must change, while accepted interventions and revised categories persist as control state for later Explorers. Together, the operators provide the expansion stage of a proposed *expansion-before-optimization* pipeline: diversify causal premises, then make the resulting candidate population cumulative and steerable.

Across eight workflows on one retirement-design problem, the released system produced 196 candidate records; its world stage comprised 25 world-and-solve branches. An unblinded post-hoc census classified bottleneck relief as clear in 17 branches and partial in eight; all 25 Solvers added multi-component architecture beyond the stipulated premise. Seven complete rejection–redirection–mechanism-change chains appeared across 75 planned graph positions. These artifacts document automated fan-out, premise-conditioned generation, and graph-conditioned redirection, but do not establish external novelty, utility, causal performance, cross-domain generality, or large-scale efficiency.

Keywords Open-Ended Problem Solving, World-Diversity Search, Counterfactual Worlds, Solution-Space Search, LLM Ideation, Multi-Agent Systems, Knowledge Graphs, Creativity Support

1 Introduction: From Answers to Solution-Space Search

Problem-directed reasoning normally asks how to find a better answer inside the causal world implied by the target. World-diversity search asks a prior question: what becomes thinkable if the system first constructs many different worlds in which the target must be solved? The unit of variation is no longer only an answer, a reasoning path, or a random cue; it is a coherent set of causal premises. At scale, the problem is therefore not obtaining another plausible proposal but deciding which world—and later which unrepresented mechanism family—the next model call should explore. We call repeated sampling from a narrow causal repertoire the *Median Trap*, an engineering failure mode whose severity varies by task, model, and intervention [1, 2, 3, 4].

The unit of search is a *candidate solution object*: a proposed intervention in an external system, represented by its mechanism, actors, operation, failure analysis, and claimed outcomes. The candidate designation separates generation from validation because originality and effectiveness are distinct requirements for creativity [5]. A problem-solving idea machine must generate mechanisms, preserve a structured account of what it has tried, and use that account to decide what kind of mechanism to request next.

The intended domain differs from bounded synthesis tasks with a fixed interface and executable score. In an open-ended social, institutional, scientific, or strategic problem, the relevant mechanism families may not be known before search, and proposals may require heterogeneous forms of evidence. The system must therefore help construct the coordinates of the solution space while it explores them. This paper instantiates and examines that architecture on one institutional-design task; applicability to other large or general problems remains a research question.

The architecture has a fan-out/fold-back geometry. World diversity provides the fan-out: separate Builders turn unrelated seeds into target-independent causal regimes, and purpose-blind Solvers attack the same target inside them. Mechanism compilation then removes the fictional surface and places every proposed intervention in a common schema. The fold-back is an ontology *of* those candidate interventions: a typed, revisable representation of mechanisms already attempted and branches that remain underrepresented. The world population expands what can become locally obvious; the candidate population makes that breadth cumulative and steerable. Here, *alien* means displaced from ordinary premises or unrepresented in the current candidate map.

Chalaco is one branch of the released 25-world fan-out. Nothing there retains its properties for more than 4.7 seconds; matter, value, and identity persist only as exchange patterns among neighbors. “Saving for retirement” is therefore a category error, but remaining woven into a durable pattern is not. The purpose-blind in-world Solver proposed pattern rights earned through long participation. Direct translation produced *Weave Collectives*, where presence earns non-transferable claims on shared resources. The example illustrates mechanism generation rather than policy validation: change the premises until an unfamiliar mechanism becomes locally ordinary, then compile its causal relation into a target-domain intervention. The candidate ontology performs the complementary task of deciding whether such a compiled mechanism repeats an occupied family and where subsequent search should go.

The machine combines breadth with memory. Many-world construction supplies multiple causal departures; ontology-guided redirection records occupied mechanism families and uses those distinctions to guide later search. The resulting architecture is *fan out, solve locally, compile, fold back, redirect, and select*. Orthogonal Insight, released in prior work on 8 December 2025, implemented the first three operations [6]; the released runner automated the world-and-solve fan-out. This paper implements the ontology map-and-redirect kernel. The graph did not yet commission new worlds, and diversity-preserving selection and progressively realistic validation remain proposed. The organizing principle is *expansion before optimization*: spend an early search budget enlarging the set of represented causal mechanisms before concentrating expensive evaluation on a small portfolio.

The research program makes two method-level claims. Orthogonal Insight contributes world-diversity search: upstream causal-regime fan-out followed by purpose-blind local solving and mechanism compilation. This paper contributes ontology-governed intervention search: a persistent, LLM-curated ontology induced from complete candidate solution objects and used as operational control state. Its curator can reorganize mechanism categories and perform diagnostic commissioning: a structural-equivalence rejection can specify which causal relation a retry must change. Accepted interventions and revised categories then condition fresh proposer contexts. Both are protocol-level contributions whose component techniques and precedents are examined in Section 5.

The eight-workflow matrix makes that grammar inspectable. It crosses three local source frames—the target alone, an unrelated word open to free association, or a complete counterfactual world and in-world solution—with three population states: none, a flat exclusion archive, or a curator-governed ontology. It supplies the richest executed combination: one displaced world-and-solution artifact plus the current candidate graph. The study contains 200 planned positions and 196 canonical candidate records—125 direct or Tabu-mediated outputs and 71 curator-accepted graph records—from one retirement-design problem and one nominal model configuration. The analysis treats the workflows as path-dependent control trajectories.

Specifically, the paper contributes:

1. world-diversity search, a released protocol for candidate-space expansion in which target-independent causal worlds supply parallel departures, purpose-blind local solves produce in-world proposals, and compilation represents their operative relations in a common target-domain schema;
2. ontology-governed intervention search, an implemented Explorer–Taxonomist controller with fresh proposal contexts, persistent typed graph memory, curator-assigned mechanism-equivalence judgments, diagnostic commissioning—rejection that specifies a missing or altered causal relation for a mechanism-changing retry—and category restructuring; and
3. an artifact-grounded study of eight path-dependent workflows, combining deterministic artifact and text diagnostics with an explicitly unblinded, post-hoc census of 25 world/Solver branches and stored population-control traces.

2 Architecture of the Solution-Space Machine

Open-ended solution search uses a branching-and-converging topology. World-diversity search branches over causal regimes; mechanism compilation converts the local solves into structured candidate interventions; a persistent ontology folds those candidates into shared control state; and progressive selection would narrow the population while protecting unusual mechanisms. This section defines the fan-out/fold-back architecture and its implementation scope.

2.1 A common design grammar: displacement, state, and context separation

The workflows form a crossed design rather than one serial pipeline. They vary two search choices: what local source frame displaces the next proposal, and what representation of the candidate population constrains it. A third choice determines which information is isolated across roles. Context is reset or withheld where carryover could encourage convergence, while explicit banks and ontologies create deliberate dependence where cumulative coverage is needed.

We call the shared role property *bounded semantic autonomy*. A proposal role is instantiated fresh and receives named, inspectable premises, memory, process, and output constraints while retaining discretion over the substantive mechanism it constructs within them. Random Inspiration leaves one unrelated word open to interpretation as a metaphor, structure, process, or principle. Semantic Tabu supplies a negative boundary while leaving the proposer to infer the mechanisms represented by prior records and invent outside them. Ontology governance assigns persistent population memory, equivalence judgment, and category revision to a curator while preserving a separate proposal role. Orthogonal Insight partitions target-blind world construction, purpose-blind local solving, and purpose-aware compilation. Autonomy is therefore local semantic discretion within explicitly supplied state.

The nine-field candidate schema provides the common output contract. It requires every proposal to expose a label, design principles, core mechanism, operation, claimed difference, rationale, failure analysis, medium-term outlook, and long-term vision, while leaving the discovery path open. Functionally, Random Inspiration supplies departure; Semantic Tabu supplies negative memory; the ontology supplies structural cartography and commissioning; Orthogonal Insight supplies causal displacement; and the hybrid cells cross source displacement with population governance. In H, ontology governance begins during transfer and admission over a fixed upstream world population. Graph-directed world commissioning is a proposed extension.

2.2 World-diversity search and the fan-out/fold-back loop

World-diversity search deliberately relocates variation upstream of the answer. It constructs K natural-language causal regimes in which conservation, agency, identity, scarcity, time, information, boundaries, or transformation may work differently, then solves the same target locally in each regime before a separate compiler sees the transfer task. The intended downstream yield is structurally displaced intervention mechanisms.

Throughout, a *world* is a generated natural-language source frame: a seed-derived root premise, a set of named consequences, and narrated implications for material processes, agency, identity, or social organization. The historical prompt called the seed a “FUNDAMENTAL LAW of physics” and instructed the Solver to treat the derived rules as operative. This wording functions as a binding instruction, not a scientific classification. Some artifacts resemble altered physical constraints—*arcual* prohibits straight trajectories and *displacement* pairs every motion with a counter-motion—whereas others literalize ontological or symbolic relations: *theca* makes containment a condition of existence, *phraseman* makes grammar causal, and *theatrical* makes witnessed dramatic tension causal. We therefore analyze the outputs as *causal regimes* or *world-rule sets*. They are not mathematical physical theories, executable environments, or simulations; internal consistency was requested in the prompt but not independently verified.

The operative analogy is the *Gedankenexperiment*. Einstein later recalled imagining pursuit of a light beam at c and reasoning from the resulting standpoint, which he described as containing an early germ of special relativity [7]. The premise was valuable as a perspective from which consequences and tensions became visible, not as an experiment one could physically perform. World-diversity search automates a population-level version of that move: a Builder stipulates a standpoint, a Solver reasons from inside it, and a compiler returns with an operative relation for the external problem. The analogy is methodological rather than evidentiary; generated worlds are not guaranteed to have the theoretical discipline or scientific value of a successful human thought experiment.

After compilation, each search item is stored as a structured intervention. In the Studio conditions, the complete nine-field record is retained as metadata on its SOLUTION node, while six fields are additionally indexed as typed facets: mechanism, outcome, design principle, criticism, justification, and claimed novelty. This makes the proposed causal structure available for comparison, criticism, and reorganization; downstream evidence is required for validation.

The full machine turns one target into a governed population. World Builder and in-world Solver calls do not require sibling-branch context and can therefore run concurrently. State dependence begins at transfer: F’s open compilers can also run concurrently, whereas G’s Tabu bank and H’s ontology make their downstream proposal paths sequential. After transfer, every branch has the same nine-field candidate form. The curator ontology can then convert the incoming stream into occupied mechanism families, underrepresented branches, and diagnostic redirections. Figure 1 separates the parallel front end, stateful fold-back, and proposed progressive evaluation.

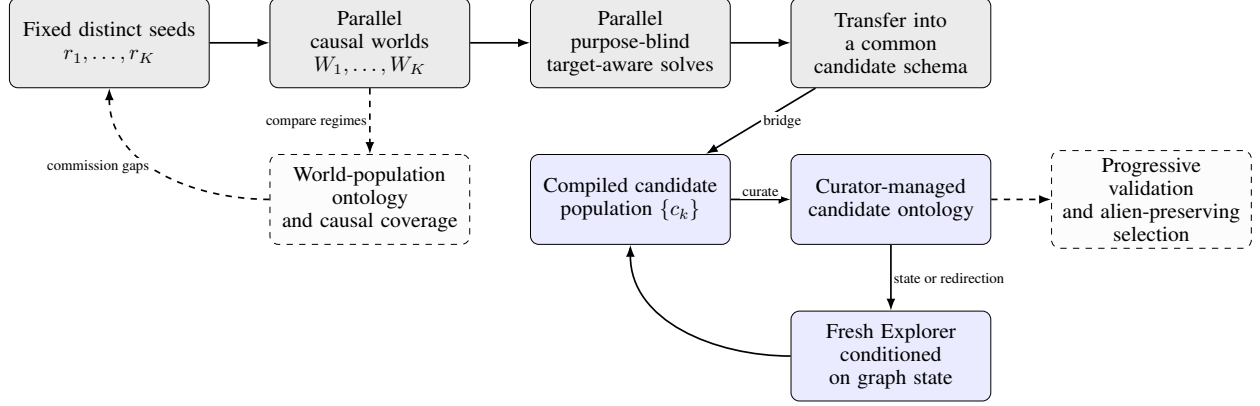


Figure 1: Many-world fan-out and ontology fold-back. The released runner generated 25 target-independent worlds and corresponding in-world solutions in five batches with phase-level parallelism, then reused those upstream artifacts across F–H. F transfer remained parallel; G and H became sequential where their accumulated bank or ontology conditioned the next proposal. In H, proposals fold into a persistent curator-managed ontology that controls admission, retries, and later Explorer context. The dashed world-population ontology, graph-directed commissioning of new worlds, and downstream selection are proposed rather than executed.

2.3 Information isolation and mechanism compilation

The proposed shortcut can be expressed as a change in the conditional distribution over generated candidates. Direct prompting samples a candidate from $p_\theta(s \mid P)$, where the ordinary framing of problem P makes familiar mechanisms highly probable. Orthogonal Insight repeats a staged transformation for branches $k = 1, \dots, K$:

$$r_k \sim \pi, \quad W_k \sim B(\cdot \mid r_k), \quad a_k \sim S(\cdot \mid P, W_k), \quad c_k \sim C(\cdot \mid W_k, a_k). \quad (1)$$

Here B constructs a target-independent world, S produces an in-world solution, and C is the open Orthogonal Insight compiler. Equation 1 describes the general operator and condition F; G and H replace C with transfer processes conditioned by an accumulating bank or graph. World construction and solving can execute without sibling-branch context, so K is a direct width control on the parallel front end. In the released run, $r_1, \dots, r_{25}$ were a fixed ordered seed set; the exact historical sampling process is not reconstructed here. A mechanism that is improbable under ordinary premises can become locally obvious under some W_k : Chalaco’s compulsory exchange makes access without accumulation an ordinary consequence rather than a request for imagination. The method changes premises until a strange mechanism becomes easy to reason about, then removes the fiction.

The implementation uses staged information isolation. A World Builder receives a random seed but not the target problem and writes a counterfactual world-rule set. A fresh Solver receives those rules and the target, but not the experimental purpose or the later transfer task; it proposes an intervention while treating the stipulated rules as operative constraints. A fresh extractor receives the source world and its in-world solution and performs *mechanism compilation*: it preserves a relational causal move while translating fictional entities into target-domain actors and operations [6].

Orthogonal Insight adds mediation to the open-ended displacement used by Random Inspiration. A target-blind Builder first expands the seed into a coherent causal constraint; a target-aware Solver then reasons inside that constraint. This separates source-world construction from target-conditioned solving.

Information is partitioned asymmetrically. The World Builder is *target-blind*: it receives the seed but not the problem its world will later mediate. The in-world Solver is *purpose-blind but target-aware*: it receives the target and world rules but is not told why the world was constructed or that a later agent will translate its answer into the target domain. The implemented worlds are textual causal thought experiments; executable simulation lies outside this implementation.

The runner launched missing World Builders concurrently within each five-branch batch, followed by concurrent in-world Solvers and, in F, concurrent open compilers. The release therefore demonstrates automated multi-world execution at $K = 25$.

This account predicts that, relative to a direct random-word cue or target-aware rule mutation, independently constructed worlds will increase the finite-budget rate of human-validated structural mechanism classes while preserving coherence. Testing that prediction requires matched-compute comparisons; the present artifacts establish the transformation itself rather than its causal advantage.

2.4 The candidate ontology as active control state

The persistent object is an ontology of complete candidate interventions. A fresh *Explorer* proposes a candidate; a recurring *Taxonomist* context evaluates it against a graph with seven node kinds—the solution container plus the six indexed facet types. In the released *Studio Model*, these nodes live in SQLite and PARENT_OF edges organize hierarchies within a type. We call the resulting controller *ontology-governed intervention search*.

Each Explorer starts in a fresh context containing an explicit rendering of the graph. The Taxonomist owns admission and graph mutation. Local proposal generation is therefore separated from the durable state that judges and redirects it, while the graph makes later proposals intentionally dependent on the accumulated population.

The controller executes the recurrent operation

$$(P, G_t) \xrightarrow{\text{graph-conditioned proposer}} s_t \xrightarrow{\text{curator}} a_t = (d_t, R_t) \longrightarrow G_{t+1}, \quad (2)$$

$$d_t \in \{\text{accept, reject+redirect}\}, \quad R_t \in \mathcal{R}^*.$$

Here P is the external problem, G_t is the current candidate ontology, s_t is a structured intervention, d_t is the admission decision, and R_t is a possibly empty sequence of category changes. A rejection can include diagnostic direction for a retry, and category changes may accompany either admission decision. Persistence resides in G_t , rather than in uninterrupted proposer or curator context. The map makes occupied and underrepresented branches available to subsequent generation as search hypotheses whose value requires downstream evaluation.

2.5 Implementation scope

The program combines three layers. The 8 December 2025 Strange Worlds release introduced world construction, local solving, mechanism compilation, the scaled generation–selection vision, and Alien Preservation [6]. This paper implements ontology-governed admission and redirection over the *compiled candidate population*. Progressive validation, recursive Solver decomposition, and ontology-guided control over the *world population* remain proposed [8].

3 Study Design and Methods

The study varies two explicit inputs to a fresh proposal role: the local source frame that displaces what can be made coherent, and the population state that constrains what later proposals should repeat, avoid, or add. It holds one open problem constant while crossing those two axes.

3.1 Task and workflow matrix

All eight workflows attacked the same open-ended problem: “How do we build a retirement system for people who don’t know how much they will earn next month, where ‘consistency’ is impossible?” The prompt supplies no enumerable answer class, executable interface, or scalar objective. The matrix holds that target constant while varying the source of displacement—none, a random word, or a counterfactual world—and the population controller—none, Semantic Tabu, or graph curation (Table 1). Twenty-five positions were planned for each executed cell, giving 200 planned slots; the target-only/no-control cell was not run.

Each cell is one path-dependent search trajectory. Tabu banks and graphs accumulate prior outputs, all cued conditions reuse the same ordered seeds, and F–H also reuse upstream worlds and alien solutions. The matrix therefore supports same-input process comparisons rather than factorial causal estimates or 25 independent replications.

Local source frame	No population state	Flat exclusion state	Curator graph state
Target only	Not run	A: Semantic Tabu	B: Studio Model
Target + unrelated cue	C: Random Inspiration	D: Cue + Tabu	E: Cue + Studio
World + in-world solve	F: Orthogonal	G: Orthogonal + Tabu	H: Orthogonal + Studio

Table 1: The 3×3 -minus-one design crosses increasing structure in the local source frame with increasing structure in externalized population state. Proposal calls or Explorer contexts are fresh; where memory exists, it is supplied through a bank or graph. The ordering describes protocol structure rather than expected performance.

3.2 Executed proposal and control paths

The rows increase source-frame structure. In C–E, a random word is a deliberately open associational cue: the prompt says not to force a literal connection and permits any metaphor, structure, process, or principle [9]. No intermediate semantic mapping is prescribed; only the completed nine-field candidate is structured. In F–H, the cue has already been expanded by a target-blind Builder into a complete world, and the local Solver output becomes part of the transfer constraint. The columns increase population-state structure. A Tabu bank tells a fresh call which structural approaches to avoid; the Studio graph shows a fresh Explorer occupied and underrepresented branches while a separate Taxonomist controls admission and category change. H combines the world-and-solution frame with graph state in one Explorer–Taxonomist transfer path. Because F–H share the complete upstream artifact, their first point of divergence is transfer.

The source population comprised 25 automatically generated world/Solver pairs from a fixed ordered seed list. The runner processed five branches per batch, with five World Builder calls concurrent within their phase and five Solver calls concurrent within the next phase. Each upstream pair was then reused across F, G, and H.

Semantic Tabu turns the accumulated archive into an exclusion prompt. Each fresh, one-turn call receives all prior structured records as JSON, reviews and lists their stated core mechanisms, declares which broader structural approaches it will avoid, and generates a new candidate. It externalizes memory but not its interpretation: every new call must infer a broader mechanism partition again from raw prior records. The method is a mechanism-level, full-record refinement of cumulative technique exclusion [10]. Its implementation operates entirely through prompt context, without a semantic index, equivalence verifier, or programmatic avoidance check.

The Studio conditions execute the paper’s central controller loop. A fresh Explorer receives the current graph state and is prompted to target an underrepresented direction; a Taxonomist role assigned to curation rather than initial proposal generation may accept, reject with mechanism-level redirection, and optionally restructure the ontology. Where Tabu externalizes raw memory, the ontology externalizes both memory and a revisable interpretation of that memory. Taxonomist sessions could restart between or within batches; the SQLite graph provided cross-batch persistence. For accepted graph candidates, each indexed field links to the closest existing node at cosine similarity ≥ 0.75 or creates a new node. Curation was qualitative; no automated similarity-rejection decision was observed in the released run.

3.3 Corpus and descriptive analysis

Runs occurred on 13–14 December 2025 with the runner configured for Claude Opus 4.5. Canonicalization recovers 196 candidate records: 125 retained top-level outputs in A, C, D, F, and G, plus 71 curator-accepted SOLUTION nodes in B, E, and H. Of the 200 run-capture files, 168 are strict JSON. The other 32 are graph-condition captures whose stored negotiation text contains unescaped model-generated JSON; for B, E, and H, the committed SQLite databases are authoritative for accepted candidates. One accepted B candidate appears only in its database. The resolved checkpoint and sampling configuration cannot be reconstructed exactly.

Exact uniqueness counts compare stripped label strings and core-mechanism strings. To describe whether a candidate lies closer to another item in its own workflow than to its nearest matches elsewhere, each core mechanism was embedded with a pinned revision of `sentence-transformers/all-MiniLM-L6-v2` using L2-normalized vectors [11, 12]. For solution i in condition c_i , we compute

$$w_i = \max_{j \neq i, c_j = c_i} x_i^\top x_j, \quad b_i = \frac{1}{7} \sum_{d \neq c_i} \max_{j: c_j = d} x_i^\top x_j. \quad (3)$$

Here w_i measures the closest other item in the same condition, while b_i averages the closest match found separately in each other condition. We report their means and ratio only as wording-sensitive descriptors; no uncertainty or causal interpretation is attached to these dependent outputs.

For each graph, we summarize the MECHANISM hierarchy by root count, maximum PARENT_OF depth, and mean child count among non-leaf nodes.

The world-diversity audit is separate from the candidate-level embedding analysis. It verifies exact-file hashes for all world and Solver artifacts, extracts the five model-labeled rules in each world, and normalizes their titles by removing an initial artifact formula such as “Law of,” lower-casing, and removing punctuation. To make the source population legible, we also assign each world’s primary causal premise to one of five broad, post-hoc author categories; these are interpretive groupings, not discovered natural kinds. Finally, a dependency-free word-TF-IDF diagnostic represents each C, F, or G candidate by its label, design principles, core mechanism, and operation. Each two-condition comparison fits a separate 50-document vocabulary using raw term counts, smoothed inverse document frequency, and L2 normalization. For F–G, it compares candidates compiled from the same upstream world-and-Solver pair with all mismatched-source pairs. A parallel C–F comparison asks whether seed identity alone leaves a comparable paired lexical signature.

A post-hoc trace census codes several preconditioning events. For each of the 25 world/Solver pairs, *bottleneck relief* is clear when a stipulated world rule makes at least one target-relevant function—such as attribution, locking, allocation, forecasting, return, eligibility, or provision—automatic, impossible, or trivial by assumption. It is partial when the premise materially eases or reframes the function without supplying it cleanly. *Downstream architecture* requires the Solver to add at least two elements among named actors, a multi-stage lifecycle, allocation or eligibility logic, and governance or failure handling. We also distinguish capabilities stated directly in the world from mixed or Solver-amplified claims, and flag cases in which the fictional operation asserts the desired endpoint rather than an upstream capability.

For A, D, and G, a history-bearing call is one after the first position; archive uptake requires one identifiable mechanism summary per earlier candidate plus an explicit exclusion plan. For B, E, and H, a hard redirect requires structural-equivalence rejection, diagnostic direction, a retry that changes the primary causal operator, and eventual admission. A non-leaf mechanism category is counted as later-visible when its short title stem—the text before an explanatory dash or colon, with terminal wrapping normalized—recurs in a later Explorer-side capture. The coding is unblinded, post-hoc, and not independently replicated; coding definitions and a summary ledger are released in `analysis/trace_coding.md`.

4 Results: Search-State Changes in the Released Artifacts

The released artifacts document a branching front end and a stateful convergence point. Twenty-five world branches instantiate automated fan-out; later traces record ontology state being supplied to subsequent proposers and curator-authored category changes being persisted for later calls. The 196 canonical records establish workflow coverage; the following results make both information flows inspectable.

4.1 Automated fan-out across 25 causal worlds

Before transfer into a target-domain candidate schema, the runner created one stored world and one stored in-world solution for each of the 25 ordered seeds. All 25 world files and all 25 Solver files are nonidentical under exact-file comparison. The prompt requested three to five named rules; every World Builder produced five, giving 125 rule instances with 123 distinct normalized titles. These counts measure artifact structure and naming variation, not physical or causal independence. No person selected a promising world before the solve stage; every branch was carried forward and reused across F–H.

The two world codes answer different questions. Bottleneck relief describes how much target-relevant work the stipulated premise supplied: 17 branches were coded clear and eight partial. Downstream architecture asks whether the Solver still had design work to do: all 25 outputs contained at least two operational elements beyond the premise. Relief was not uniformly disciplined, however; on the separate final-outcome-fiat test, 14 passed, eight were borderline, and three failed by asserting the desired endpoint too directly.

Table 2 makes the breadth concrete. The worlds span causal premises involving irreversible material joins, compulsory exchange, relational boundaries, rotational geometry, witnessed action, grammar, need, and other regimes. Their compiled mechanisms likewise range from hard time locks and automatic effort credits to presence-based resource rights and witnessed contribution claims.

Primary causal family	<i>n</i>	Representative world premise	Representative F compilation
Material transformation, persistence, or enclosure	7	<i>limelike</i> : unlike matter fuses irreversibly	Cryptographic time locks make each contribution inaccessible before retirement.
Flow, exchange, or gradients	6	<i>chalaco</i> : properties circulate every 4.7 seconds	Long participation earns presence-based claims on a continuing resource flow.
Relational or topological causation	3	<i>paranucleic</i> : boundaries determine what an interior can be	Security depends on embeddedness in reciprocal care relations rather than one balance.
Motion geometry as existence	3	<i>arcual</i> : all trajectories curve and tend toward return	Retirement tracks lifetime trajectory capture rather than monthly timing.
Symbolic, social, or mental causation	6	<i>theatrical</i> : witnessed acts acquire causal force	Publicly witnessed sacrifice ratios create claims on pooled resources.

Table 2: A post-hoc author grouping of the 25 released source worlds. Counts describe one interpretation of each primary premise; they do not establish exhaustive or objectively independent world categories.

The common source left a detectable downstream signature. F and G candidates compiled from the same world-and-Solver pair had mean word-TF-IDF similarity 0.189, compared with 0.067 across mismatched source pairs, a ratio of 2.83. By contrast, direct random-word C and world-mediated F candidates sharing a seed were no more similar than mismatched seeds (0.076 versus 0.075). This is consistent with the shared world-and-Solver source leaving a larger lexical imprint than seed identity alone; it is not evidence that the resulting mechanisms are causally distinct, creative, or useful.

The same audit also exposes convergence. Several worlds occupy overlapping families, and radically different premises can compile back into hard commitments, pooling, automatic capture, participation-based claims, or social witnessing. The release establishes automated world generation at width 25; causal-space coverage, marginal mechanism yield, and selection efficiency remain open evaluation targets.

4.2 When the ontology changes the next proposal

Run E20 makes population control visible. With nineteen accepted candidates already represented, the Explorer proposed *Paranucleic Savings Satellites*: monitor personal spending rather than income, infer capacity, and trigger retirement contributions. The Taxonomist classified the proposal as the already occupied pattern “individual monitoring → automated trigger” and explicitly tested whether changing the input signal changed the structural category. The rejection then commissioned four alternatives: redistribute value to others, operate only on aggregate signals, route ownership by purchase category, or invert the trigger.

The Explorer chose category-directed routing and returned *Consumption-Mirrored Ownership Routing*. Instead of estimating saving capacity, purchase categories routed flat micro-contributions to sectoral equity—groceries to food, rideshares to transport. The Taxonomist accepted the path after comparing it with existing asset-transmutation mechanisms because its operating relation had changed from detection to routing. Thus, graph categories supported equivalence judgment, targeted commissioning, and mechanism revision. Some retries selected Taxonomist-named alternatives, demonstrating responsiveness to redirection rather than independent discovery.

Run H16 shows the complementary operation. *Earning Pattern Compliance System* measured contributions against a worker’s personal earning cycle rather than a universal schedule. The Taxonomist judged that the proposal did not fit its existing top-level stances toward inconsistency—accept, eliminate, leverage, obscure, validate, distribute, or transform. It created a new mechanism root, *Reframe Inconsistency*, for the claim that apparent irregularity may be “consistency to a different rhythm,” and then added the candidate. That distinction became part of the graph presented to later Explorers.

E10 shows a third control path. The Explorer’s cue-triggered payment proposal renamed an occupied real-time income-detection mechanism. The curator rejected it and commissioned a design with no automated detection, payment processor, or digital infrastructure. The retry produced *Benefit Night Labor Troupes*: labor obligations flow to an ensemble rather than accumulating as individually tracked monetary contributions. The selected sequences sit within the broader trace census in Table 3. The table keeps unlike denominators separate: a world/Solver branch, a history-bearing Tabu call, a planned graph position, and a non-leaf mechanism category are different observational units.

The 72 eligible Tabu calls were the 24 post-first positions in each of A, D, and G. Together, they restated earlier candidates 900 times, counting repeated exposure across calls, and every call declared an exclusion plan. In 18 calls,

Trace phenomenon	Applicable unit	Observed	Descriptive interpretation
World premise removes or materially eases a target bottleneck	25 world/Solver branches	17 clear; 8 partial	Under the rubric, all 25 branches were coded clear or partial; relief strength remains interpretive.
Solver adds multi-component structure downstream of the premise	25 world/Solver branches	25/25	All 25 outputs were coded as containing at least two downstream architectural elements.
Capability provenance in the stored world	25 world/Solver branches	8 direct; 14 mixed; 3 amplified	Fourteen mixed an explicit world capability with Solver extrapolation; three depended primarily on the latter.
Final-outcome fiat	25 world/Solver branches	14 pass; 8 borderline; 3 fail	Three Solver derivations asserted the endpoint directly; eight were coded borderline.
Archive inventory plus explicit structural-exclusion plan	72 history-bearing calls (24 each in A, D, and G)	72/72	All eligible traces restated prior mechanisms and declared exclusions; semantic non-equivalence remains untested.
Structural rejection → diagnostic direction → mechanism-changing retry	75 planned graph positions; 71 admissions	7 complete chains	Seven admitted retries met the full criterion; the graph’s causal contribution is not isolated.
Non-leaf mechanism category named again later	46 final non-leaf categories	40/46	Most category vocabulary remained visible to a later Explorer; causal use is not isolated.

Table 3: Post-hoc census of trace-level preconditioning phenomena. Counts are descriptive readings of stored artifacts, not independent creativity or quality judgments.

the number of prohibition bullets was smaller than the number of prior records; this may reflect grouping or omission, neither of which was semantically evaluated.

Across traces for 75 planned graph positions, seven complete hard-redirect chains ended in canonical admission. The other 64 of 71 admissions did not contain a complete chain in the bounded captures, and the four no-admission wrappers were not separate semantic rejections. Forty of 46 non-leaf mechanism-category titles recur in a later Explorer-side capture, showing durable availability rather than causal use. Five additional curator directions match later admissions, but asynchronous pane capture prevents clean before/after attribution. A causal creativity comparison would require shuffled-archive, fixed-taxonomy, and matched single-agent controls.

4.3 When an alien world makes another mechanism ordinary

Chalaco illustrates the complete staged source-to-target transformation. Every property flows to a neighboring entity after roughly 4.7 seconds, isolation is lethal, and only exchange patterns can persist. The Solver therefore calls saving a category error and derives pattern rights from long participation in intergenerational exchange knots. F’s direct bridge compiles this into *Weave Collectives: Presence-Based Security*: small groups in which continued participation earns non-transferable claims on housing, care, food, and governance.

A matched-source pair from the first run shows that the same operator can also converge. In the *limelike* world, deposited matter accretes irreversibly and cannot be separated before a life-stage trigger. F compiles the shared world and alien solution into *Cryptographic Accretion Protocol*; H compiles them into *Irrevocable Accretion Contracts*. Both use time-locked contracts to make early withdrawal structurally impossible. Because H’s graph was empty for its first proposal, no occupied family was available for redirection. The pair therefore illustrates a boundary case for the controller.

These traces concern mechanism formation. Whether the policies work, and whether matched direct prompting reaches the same mechanisms, remain open evaluation questions. The *desperacy* trace supplies a second inversion—income volatility changes from a savings liability into a coordination credential—and Appendix A follows its world rule, *Resonance Banking*, and extracted *Resonance Pods*.

4.4 One seed, six search trajectories

Holding the cue *theatrical* fixed did not produce a common mechanism. Six workflows retained traces of theatrical language while routing it toward intervention structures ranging from crisis-fund balances to witnessed reciprocal obligations (Table 4).

4.5 Corpus-scale distinctness—and its limit

Exact-string uniqueness was complete, but semantic separation was modest. All 196 labels and core-mechanism strings were unique under exact comparison; mean closest-match similarity was 0.563 within conditions and 0.514 across conditions, a ratio of 1.10 (Table 5). D and E had the largest ratios, while F and G were closest to parity. These diagnostics rule out literal duplication, not functional equivalence, and may reflect condition-specific wording as readily as mechanism concentration.

Condition	Generated label	Mechanism emphasis
C	The Dark Theater Fund	A working-life crisis fund whose unused balance later becomes retirement income.
D	Standing Cast Retirement	Platforms fund maintained labor readiness, including periods without deployment.
E	Benefit Night Labor Troupes	Members owe benefit labor to a troupe that later supplies care and maintenance.
F	Witnessed Sacrifice	Claims depend on contribution as a share of current income, combined with public witnessing.
G	Retirement	Recorded testimony and hardship context weight contribution claims.
H	Testimony Accord	Witnesses stake reputation and acquire reciprocal support obligations.
	Witnessed Covenant Mutual Obligation	

Table 4: Six routes from the same released seed. Mechanism emphases are author summaries; functional distinctness was not independently coded.

Condition	n	Within $\bar{w}$	Between $\bar{b}$	Ratio
A: Semantic Tabu	25	0.573	0.527	1.09
B: Studio Model	23	0.585	0.521	1.12
C: Random Seed	25	0.575	0.537	1.07
D: Seed + Tabu	25	0.656	0.569	1.15
E: Seed + Studio	25	0.566	0.484	1.17
F: Orthogonal	25	0.524	0.499	1.05
G: Orthogonal + Tabu	25	0.496	0.486	1.02
H: Orthogonal + Studio	23	0.532	0.484	1.10
All records	196	0.563	0.514	1.10

Table 5: Best-match cosine similarities under Equation 3. No uncertainty estimate or causal interpretation is attached to these dependent outputs.

4.6 The final candidate ontologies

Three sequential searches produced visibly different mechanism hierarchies. E created 49 mechanism nodes—40% more than B—and reached depth 3; B and H each ended with 10 roots, while H had the highest mean branching. None of the final graphs contains dangling edges, cycles, or mechanism nodes with multiple parents (Table 6).

Condition	Solutions	Mechanisms	Roots	Depth	Branching	All nodes
B: Studio Model	23	35	10	2	1.92	167
E: Seed + Studio	25	49	5	3	2.00	197
H: Orthogonal + Studio	23	34	10	2	2.18	169

Table 6: Final curator-authored graph statistics. Root count, depth, and branching refer only to the stored MECHANISM hierarchy; “All nodes” includes all seven node types.

Counts conceal the category structures that produced them. Figures 2–4 visualize the mechanism hierarchies for B, E, and H. They show the curator’s conceptual backbone rather than every stored relation: solution nodes and their USES_MECHANISM links are omitted, synthetic super-roots make independent categories visually comparable, and compressed boxes summarize lower mechanism branches.

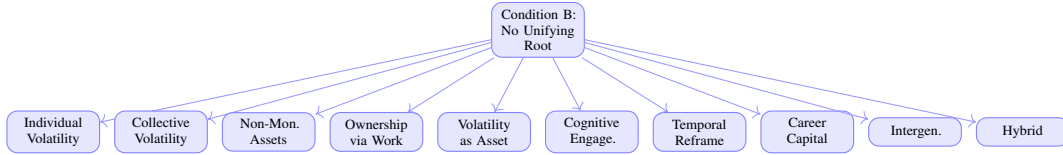


Figure 2: Condition B mechanism-tree schematic. Ten independent mechanism roots are shown beneath a synthetic super-root; solution nodes, candidate-to-mechanism links, and lower mechanism branches are omitted. The stored hierarchy organizes proposals around domain concepts including volatility, non-monetary assets, work-based ownership, and temporal reframing.

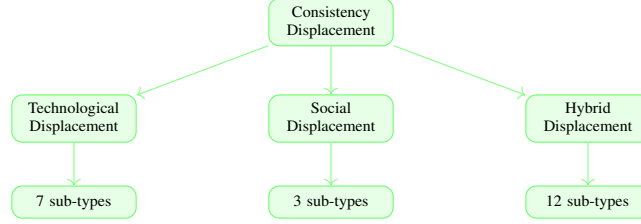


Figure 3: Condition E mechanism-tree schematic. The dominant curated backbone divides Consistency Displacement into technological, social, and hybrid branches; compressed boxes summarize lower mechanism categories, while solution nodes and candidate-to-mechanism links are omitted. Four additional candidate-text mechanism roots retained in the database are not shown, yielding the five-root total in Table 6.

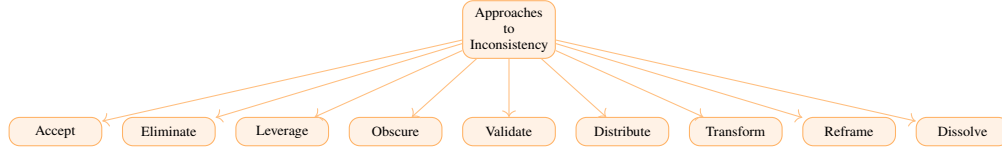


Figure 4: Condition H mechanism-tree schematic. Nine curator-authored stances toward inconsistency are shown beneath a synthetic super-root; solution nodes, candidate-to-mechanism links, and lower mechanism branches are omitted. One additional candidate-text mechanism root retained in the database is not shown, yielding the ten-root total in Table 6.

The schematics expose three different cartographies of the same task. B partitions proposals through domain concepts such as volatility, non-monetary assets, ownership, and time. E places most mechanisms beneath one Consistency Displacement doctrine and separates technological, social, and hybrid realizations. H instead organizes meta-strategies toward inconsistency itself—accept it, leverage it, transform it, reframe it, dissolve it, and related stances. These are curator-produced search representations, not discovered natural kinds.

These path-dependent histories describe curator-produced search state. With one trajectory per condition, topology cannot be attributed separately to source, order, prompt, or linking policy. The curator nevertheless constructed and revised a typed representation while search was underway.

Read descriptively, admission order is also inspectable. B’s early admissions cluster in contribution mechanics (surge capture, rotating circles, volatility-hedging pairs, smoothing buffers, time banking) and then market instruments (volatility derivatives, prepaid consumption rights, platform equity); later positions broaden and intermittently reframe rather than climb monotonically. Run 9’s *Anti-Fragile Retirement Accounts* treats volatility as a gain source, runs 16–17 shift to non-monetary stores (reputation equity, generational apprenticeship bonds), financial designs recur through the final third, and run 24’s *Counter-Cyclical Retirement Pool* reverses flow direction so the pool pays the worker in low months. The captured negotiation at the final planned position proposes a “Pattern-Learning Retirement Autopilot” that shifts per-contribution timing decisions from the worker to the system; no canonical record was admitted there. This is a post-hoc reading of one prompted, path-dependent admission order; it documents broadening and intermittent reframing, not category exhaustion or an abstraction hierarchy.

5 Related Search Architectures

Fictional worlds and displaced source spaces have long been used to stimulate creativity. Arnold’s 1953 *Arcturus IV* case placed human designers inside one alien planetary and biological setting so that ordinary product assumptions no longer applied; Random Entry, Synectics, design fiction, and analogical prompting likewise interrupt fixation [13, 9, 14, 15, 16, 17, 18]. The present protocol extends this tradition into an automated search operator over a *population* of newly constructed causal source worlds, with target isolation during construction and explicit compilation back to one external problem.

Direct lexical stimulation supplies a second component precedent. Mehrotra et al. prompt a model to incorporate a random object or one of its attributes into product designs, stories, and marketing artifacts [19]. MetaMuse uses four random dictionary keywords as external stimuli and develops them through fixed waypoints—property extraction, problem mapping, complete solution formulation, and code generation [20]. Condition C instantiates an unscaffolded

Random Entry baseline: one word may become any metaphor, structure, process, or principle, while the nine-field schema constrains only the completed intervention. Its free-association policy differs from scaffolded analogical mapping, while remaining within the established lexical-stimulation family.

The task scope also differs. MetaMuse searches executable cache-replacement and bin-packing algorithms inside fixed interfaces and measurable feedback spaces. The present study searches an institutional mechanism space with no predefined solution classes or automatic score, making representation and qualitative population control central. Universe of Thoughts also treats ill-defined, open-ended problems and transforms target-derived rules or analogous spaces [21]. World-diversity search differs operationally by interposing a population of target-independent causal worlds, purpose-blind local solves, and a separate mechanism-compilation stage.

Semantic exclusion likewise has direct precedents. Denial Prompting detects techniques in earlier solutions and accumulates prohibitions; NoveltyBench conditions later answers on earlier ones; G2 summarizes approaches in previous responses and requests a solution using different methods or concepts [10, 3, 22]. Semantic Tabu is a mechanism-level, full-record variant: each fresh call receives the structured intervention archive, states broader approaches to avoid, and then proposes another complete intervention.

Automated many-environment expansion also predates this work. Enhanced POET generates an evolving population of learning challenges, filters new environments for meaningful novelty, and transfers policies among them [23]. That world-level diversity control is an important precursor to the future system proposed here. Its environments are tasks to be mastered by embodied policies, whereas world-diversity search uses target-independent natural-language causal regimes temporarily imposed on one external semantic problem and then compiled into target-domain intervention mechanisms. MiroFish constructs graph-grounded parallel social worlds from supplied source material and simulates persona agents for prediction and scenario interrogation; its graph represents actors and event memory in the simulated society rather than a candidate population used to redirect solution proposals [24].

Population expansion, archiving, and selection are also established. Novelty search and MAP-Elites maintain diverse repertoires, while QDAIF combines language-model proposal and evaluation with an evolving MAP-Elites archive [25, 26, 27]. QDAIF’s main studies use researcher-chosen diversity dimensions; Appendix A.10 of Bradley et al. adds a preselected second dimension during search and proposes automatic generation of new axes. AI Co-Scientist generates, reflects on, clusters, ranks, and evolves scientific hypotheses; its Proximity graph supports clustering and de-duplication, while separate research overviews and Meta-review feedback direct Generation toward unexplored areas [28]. Together these systems establish generate–organize–select and population-feedback search. The present controller concentrates admission, diagnostic redirection, and category revision in one dynamically restructurable ontology of complete interventions that governs fresh proposal contexts.

ConceptVis, IdeaHound, and Agentic Deep Graph Reasoning already make conceptual search cumulative. ConceptVis is a human-steered product-concept and design-knowledge branching interface: designers expand selected nodes to control breadth and depth, but it does not induce a curator-owned ontology of abstract causal mechanisms across complete interventions. IdeaHound exposes gaps in an emerging semantic solution map; Agentic Deep Graph Reasoning generates prompts from underexplored regions of an evolving domain concept–relation graph [29, 30, 31]. These are genuine precedents for graph-guided exploration. The distinction here is the representational object and automated control path: recurring LLM curator contexts act on a persistent typed ontology of complete problem-specific intervention records, accept or reject with diagnostic resubmission, reorganize mechanism categories, and return the changed state to fresh proposers.

Two typed-graph systems provide close representational comparisons. SciAgents combines specialist agents with a literature-derived ontological knowledge graph and emits structured scientific hypotheses containing mechanisms, outcomes, and design principles; generated hypotheses do not become the population that rewrites and steers that graph [32]. Massoudi and Fuge’s Design-State Graph is an evolving typed artifact refined by a nine-role agent system, but it decomposes one engineering design into requirements, embodiments, and models rather than maintaining an ontology of many alternative solutions whose occupancy governs subsequent search [33]. These systems implement multi-agent typed-graph reasoning at different representational levels.

This population representation also differs from a graph of reasoning steps. Chain-of-Thought externalizes intermediate reasoning toward one answer; Tree of Thoughts and Graph of Thoughts branch, evaluate, combine, or backtrack over intermediate states [34, 35, 36]. Ontology-governed search persists across complete candidate interventions and changes the categories shown to and used to judge later proposers.

To our knowledge, two protocol-level contributions remain distinct from the precedents above. First, among systems available before the 8 December 2025 Orthogonal Insight release, we found no earlier report of *world-diversity search as specified here*: a population of target-independent textual causal-world constructions, purpose-blind local solves of one external target under their stipulated rules, and separate compilation of the operative relations into target-domain

interventions. Second, among systems available before the original 7 March 2026 release of this paper, we found no earlier report of ontology-governed intervention search as implemented here: a persistent typed ontology induced from complete candidate interventions, fresh graph-conditioned proposer contexts, and recurring LLM curation through mechanism-equivalence judgment, acceptance, diagnostic commissioning through rejection and resubmission, and category restructuring. These claims concern the specified protocols; their component techniques are the established precedents summarized above.

6 Discussion

The artifacts support two implemented motions with a broader design implication. They show world diversity being used upstream of solution generation and an evolving candidate representation re-entering later generation. Together these motivate, but do not yet validate, a machine that governs both which causal worlds are explored and which compiled mechanisms receive further search and evidence.

6.1 The map becomes part of the generator

The graph served as operational control state in two observable senses. Forty of 46 non-leaf mechanism-category titles recur in a later Explorer-side capture, while seven complete hard-redirect chains ended in admission across 75 planned graph positions. The first count records durable vocabulary and the second selective intervention; neither identifies a benefit. The E20 retry and H16 root creation illustrate how the candidate representation could re-enter generation and change while search was underway. Isolating the effects of graph visibility, curator prompting, role separation, extra turns, and qualitative selection requires further controls. A curator-side control could ask whether the operation types themselves—rejection with direction, category creation, and restructuring—arise under a prompt that requests only a legible taxonomy without naming them. Such a control would test whether the operations arise unprompted, not whether they improve search.

A fixed archive asks where a candidate belongs within an existing partition. A co-evolving ontology can also ask whether the partition expresses useful distinctions, whether apparently different branches share one mechanism, and whether a proposal requires a new category. Sparse roots can commission targeted proposals; uncombined mechanism–outcome pairs can become search prompts; repeated rejection without new nodes can inform a saturation diagnostic. The ontology is therefore a revisable hypothesis about mechanism space—not a neutral map of all possible solutions.

Viewed at the system level, this loop implements a limited operational form of *metacognition*: the machine monitors its own search history and uses that representation to control subsequent search [37]. The ontology is an externalized model of what the machine has proposed, which mechanism families it currently recognizes, where it judges proposals equivalent, and which regions it regards as underrepresented. The Taxonomist performs monitoring and control by interpreting the population, revising its categories, and redirecting generation; the Explorer performs the next object-level generation step. The metacognitive unit is therefore the composite Taxonomist–ontology–Explorer machine rather than an isolated language-model call. The released traces establish this information flow; its effect on search quality remains to be measured.

6.2 Preconditioning generation while preserving local autonomy

The workflows share a design pattern without performing the same operation: each changes some part of the state from which a complete candidate is generated. Random Inspiration supplies an exogenous association. Semantic Tabu asks the proposer to treat represented approaches as unavailable. Orthogonal Insight changes the causal premises of the local solve. The proposed oracle-premise operator would mark one chosen bottleneck as solved. Ontology governance instead changes the population’s coordinates by deciding which mechanisms are equivalent, which distinctions deserve categories, and which regions should be commissioned. Across these workflows, part of the search work moves from improvisation inside one answer to an explicit state transformation before the next proposal.

Diagnostic commissioning is the most population-specific operation in this sequence. Rather than supplying a generic cue, exclusion, or premise, the Taxonomist diagnoses structural repetition or absence in the current candidate population and converts that diagnosis into constraints for the next proposal. In E20, the retry instantiated one of four Taxonomist-named alternatives; the trace therefore establishes specification and responsiveness, not independent discovery or benefit.

These operators condition generation at different levels. World construction, and the proposed oracle-premise extension, alters the premises of a local solve. Graph restructuring leaves those premises unchanged and alters the categories and gaps explicitly supplied to later calls. H16 added *consistency to a different rhythm* as a persistent category; that

category appeared in H17’s context, after which H17 repeated its personal-rhythm mechanism and was redirected to the thin *Obscure Inconsistency* branch. E20 records a corresponding retry from signal detection to ownership routing. These temporal sequences are consistent with representation-sensitive redirection but do not identify the graph’s causal contribution.

The architecture combines contextual independence with deliberate population dependence. H supplies the richest executed crossing: a fresh Explorer receives a world-and-solution artifact plus the current graph, while the Taxonomist controls admission and map revision. This is bounded semantic autonomy: local generation has discretion within inspectable external constraints. Externalizing category formation, structural-equivalence judgment, gap detection, and commissioning makes them explicit population-level operations across fresh contexts. This differs from architectures that organize intermediate reasoning states toward one current answer; the relevant distinction is the scope of the search state, not the visual form of a graph or chain.

6.3 World diversity should itself become governed search state

Each released World Builder operated independently from its siblings, and diversity control began only after candidate compilation. The 125 named world rules and five broad families in Table 2 make the realized variation inspectable inside a fixed template, but causal overlap among source worlds remained unmanaged. All 25 Solver outputs contained multi-component downstream architecture, yet three derivations failed and eight were borderline on the final-outcome-flat test. Automated width therefore produced usable reasoning traces under the codebook without guaranteeing disciplined causal regimes. A governed world population would distinguish seed diversity from causal-regime diversity.

A next system could maintain a *world-population ontology* alongside the candidate-solution ontology. Before the target is disclosed, each world could be represented by a causal signature over temporality, conservation, agency, identity, scarcity, information flow, boundary structure, permitted transformations, and social or physical enforcement. A World Taxonomist could merge near-equivalent regimes, split inadequate categories, reject shallow restatements, and commission Builders into thin or unrepresented regions. This preserves target blindness while making the front end a coverage-seeking search rather than a collection of unrelated one-shot generations.

A governed world generator could combine random words with stipulated physical constraints, evolutionary or ecological histories, technological path dependencies, historical branch points, institutional or cultural inversions, and systematic removals or substitutions of familiar capabilities. Operators could also compose departures, such as alternative evolution inside a resource regime with altered conservation rules. Their utility is an empirical question; explicit causal signatures would make the resulting world diversity and downstream mechanism yield testable. The prior release proposed a world-richness ablation [6]: whether sparser or richer textual elaboration than the released worlds changes downstream mechanism variety when the seed, model, and solve protocol are held fixed. Executable simulation is a separate dimension from textual elaboration, not its endpoint.

A separate extension would deliberately relax target blindness. A target-aware, single-property *oracle-premise* operator would stipulate that one known part of a problem is already solved, then derive a more complete solution conditional on that premise. In a cancer-research thought experiment, for example, the world might stipulate that malignant and healthy cells are perfectly distinguishable. Solving locally could expose the remaining requirements for delivery, timing, resistance, monitoring, and combination; compilation would then work backward, treating perfect discrimination as a design specification and asking which real biological or technological mechanisms could approximate it. None of the released runs implemented this operator because every Builder was target-blind. They nevertheless contain incidental analogues: in *phosis*, an unavoidable gradient made contribution capture automatic before the Solver derived pooling, storage, reserves, and verification; in *limelike*, irreversible binding made commitment a given before the Solver derived variable intake and staged release. These traces motivate rather than instantiate oracle-premise search.

Oracle-premise decomposition is directed and potentially cheaper because a person or agent chooses the bottleneck in advance. Target-blind strange worlds search more broadly: they may expose premises and relations that nobody knew to stipulate. A governed portfolio could allocate effort across both. Oracle-premise worlds would generate hypotheses rather than biological simulations or evidence of efficacy; resulting mechanisms would require domain-expert review and empirical validation.

World novelty and productive novelty must then be evaluated separately. *Ex-ante world diversity* asks whether a source regime adds an unrepresented causal structure before the target is known. *Ex-post mechanism yield* asks whether solving and compiling that regime adds an intervention mechanism absent from the candidate ontology. Repeated seeds, matched compute, blinded causal coding, and marginal-yield curves could test whether governed world coverage discovers more portable and useful mechanisms than random words alone. The world ontology would decide where to look; the candidate ontology would decide what the search found.

6.4 How fan-out and fold-back spend a finite budget

Many-world fan-out spends budget on additional premise-level branches, whereas ontology fold-back spends it on maintaining, comparing, and redirecting the compiled candidate population. A random cue perturbs the answer context, whereas Orthogonal Insight first perturbs the causal world in which an answer must make sense. A detour through an alien world may therefore become a shortcut through the model’s ordinary proposal distribution.

The architectural scaling hypothesis has two parts. Because world construction and solving do not require sibling-branch context, width can be increased and parallelized before stateful transfer; open F compilation is parallelizable as well. Because G and H use accumulating control state, however, their current transfer paths are sequential. A future world ontology would add its own coordination cost before solving, while the candidate ontology moves the later bottleneck toward mechanism-equivalence judgment, graph maintenance, and validation. A suitable evaluation measures the number of distinct, useful mechanisms per token or model call as world coverage and candidate-graph size grow. The present study does not estimate marginal yield per added world, curator cost, or end-to-end efficiency.

6.5 From search kernel to a problem-solving machine

The release demonstrates the first scaling motion at modest width. Beyond that experiment, we propose a machine that expands solution space deliberately. More target-blind World Builders could construct additional causally coherent worlds; purpose-blind in-world Solvers could work on the target without knowing the experiment’s purpose or later transfer task; and fresh extractors could compile their operative relations into candidate interventions. A persistent curator ontology would normalize mechanism descriptions, reject aliases, restructure inadequate categories, and commission proposals into thin or uncombined branches. The scaling objective is the number of structurally differentiated and inspectable mechanisms obtained under a finite compute budget.

Scale decomposes into four operational axes: raw world width K , governed causal coverage of the world population, breadth $|G_t|$ of the compiled candidate ontology, and evidentiary depth allocated to survivors. The release exercises raw width and candidate breadth at modest scale; world-level coverage control and progressive evidentiary depth remain proposed. Large-scale execution would also require concurrent graph-update policies, curator calibration, and stopping rules.

A complete problem-solving pipeline additionally requires selection and evidence. Candidates would first face inexpensive criticism and mechanism-equivalence checks; survivors would receive progressively more detail through domain models, adversarial comparison, increasingly realistic simulation, literature and data review, expert judgment, and—where safe—real-world testing. The prior release proposed portable, inverse, and magical labels for compiled mechanisms [6]; read as nonexclusive, time-stamped transfer annotations, they would let a population record when a mechanism that was magical under earlier technology has become portable. They were not applied in this experiment. If evaluation criteria are themselves typed nodes, graph structure can prioritize which candidates receive evaluation attention next; connectivity can direct scrutiny, but it cannot itself establish value or feasibility, which require the downstream evidence above.

Selection must preserve structural diversity while candidates mature. One hypothesized failure mode is prior inheritance: a Selector drawn from the same model family may reproduce correlated familiarity preferences, so unguarded best-of ranking risks prematurely down-ranking displaced mechanisms. This risk is stated as a design consideration, not a measured effect. Quality-diversity methods motivate preserving niches rather than collapsing immediately to one global score [25, 26, 27]. *Alien Preservation* names the corresponding requirement here: an unusual branch is given enough development and evidentiary testing before familiarity or early feasibility judgments may eliminate it [6]. The stage’s output format is a further design choice, carried over from the prior release [6]: a single best mechanism, a top- k structurally diverse portfolio for human review, or a weighted survivor set that preserves uncertainty rather than collapsing it—calling those weights probabilities would require calibration. For a medical problem, the ontology could allocate attention across intervention mechanisms and claimed outcomes while scientific evidence determines which candidates survive. In such a system, the graph would govern where evaluation effort goes without substituting for efficacy or safety. Whether this architecture discovers more useful mechanisms per unit of compute remains the central experiment.

Speculative extensions. The prior release and this paper’s original version sketched three further directions, retained here as unevaluated design hypotheses [6]: cross-domain instantiation of the same operators (planning, argumentation, forecasting, red-teaming); world-specialized models fine-tuned on corpora generated within one world-rule regime, intended to approximate different counterfactual causal structures; and distributed operation, in which many parties submit problems and share curated candidate populations. The parallel front end is amenable to distribution; the stateful

fold-back is not directly distributable and would depend on the concurrent graph-update policies noted above. None of these is implemented or evaluated here.

7 Evidence Boundary and Limitations

Table 7 distinguishes implemented components and descriptive artifact analyses from hypotheses concerning creativity, utility, causal advantage, and large-scale efficiency.

Design and dependence. The design cannot isolate architecture effects. It lacks a contemporary direct/no-control cell; banks and graphs accumulate in fixed order; F–H share upstream artifacts; and ontology conditions jointly add role separation, graph visibility, persistent state, extra turns, tool use, rejection, and accepted-only selection. The 196 records are therefore dependent outputs, not independent experimental replications.

Construct validity. Exact-string and embedding differences do not establish mechanism-level novelty or quality. Schema pressure and condition-specific language may influence both, while graph structure reflects curator policy as well as candidate structure. The trace census uses one unblinded, post-hoc coding pass without an independent second coder. No human mechanism-equivalence panel, domain-expert assessment, or outcome evaluation was performed.

Scope. The study covers one retirement prompt, one nominal model configuration, and one trajectory for each ontology condition. It includes no varied proposal orders, held-out seeds, alternative models, replicated graphs, or compute-matched ablations. Transfer to other domains remains open.

Claim	Status	Evidence boundary
Eight workflows; 196 canonical records	Implemented, audited	125 retained outputs plus 71 curator-accepted graph records; one prompt, one nominal model configuration, dependent sequential outputs.
Automated 25-world fan-out	Implemented, audited	25 stored world texts and 25 Solver outputs; phase-level parallelism in five-branch batches; no width sweep or throughput benchmark.
World-rule, premise-relief, and shared-source diagnostics	Post-hoc descriptive coding	125 model-labeled rules; 17 clear and 8 partial bottleneck-relief codes; 25/25 downstream-architecture codes; 14/8/3 pass–borderline–fail on final-outcome fiat; an F–G lexical signature; no independent coder.
Exact-string uniqueness and embedding ratios	Descriptively evaluated	Text properties under one encoder; no functional-equivalence judgment.
Three graph topologies	Descriptively evaluated	One final curator-authored graph per graph condition.
Tabu archive uptake	Post-hoc trace-coded	72/72 later responses restated prior mechanisms and declared exclusions; semantic avoidance was not evaluated.
Graph redirection and category reuse	Post-hoc trace-coded	Seven complete chains across 75 planned positions, ending in 7/71 admissions; 40/46 non-leaf category titles recurred later; no causal ablation.
Candidate ontology used as control state	Implemented	Graph state was supplied to every Explorer; selected traces explicitly used it in rejection, retry, and category mutation. Benefit was not isolated.
World-mediated probability uplift or faster discovery	Hypothesized	No matched-compute or causally identified direct-cue comparison, independent mechanism-equivalence coding, or normalized hit-rate estimate.
External novelty, quality, usefulness	Not evaluated	No domain experts, blind judges, ground truth, adoption, or outcome study.
Causal superiority of graph or multi-agent roles	Not identified	Conditions also differ in memory, context, turns, tool use, compute, and selection.
Larger-scale efficiency or coverage growth	Not evaluated	No cost curve, saturation study, or mechanism-yield estimate as K and $ G_t $ grow.
World-population ontology and commissioning	Proposed	No world archive, causal-equivalence gate, adaptive world categories, or graph-directed World Builder ran.
Stopping rules and graph-based selection	Proposed	Code exposes some graph queries; downstream benefits were not tested.

Table 7: Claim-status map for the released study and later proposals.

8 Conclusion

World-diversity search changes the object being diversified. Instead of asking for many answers inside one assumed causal frame, it constructs many textual causal frames in which different mechanisms can become ordinary, solves locally inside them, and compiles the resulting relations into interventions for one external problem. The released operator performs this transformation across 25 worlds; ontology-governed intervention search supplies the fold-back. Fresh, graph-conditioned Explorers propose complete candidate interventions, while recurring Taxonomist contexts read and revise a persistent ontology, issue mechanism-equivalence judgments, and turn some structural rejections into targeted commissions for retries or later generation. Taken as one machine, the Taxonomist, ontology, and Explorer form an externalized metacognitive layer over generation: the system represents its own search history and uses that representation to decide where thought should go next.

The trace census extends this process evidence beyond individual examples. Premise-conditioned generation appears throughout the world corpus, while complete ontology rejection–redirection chains occur selectively. The artifacts document premise-conditioned generation and graph-conditioned redirection; they do not establish that either operation improves novelty, quality, or usefulness.

The common design principle is bounded semantic autonomy. Fresh generative roles reason richly inside explicit local frames, while long-lived coordination resides in inspectable artifacts outside those roles. Random cues and causal worlds occupy progressively richer positions on the source-frame axis; Tabu banks and candidate ontologies do the same on the population-state axis. This architecture is intended for open-ended problems in which neither the important solution families nor a universal evaluator are supplied in advance. The released study documents this use on one institutional-design problem; cross-domain generalization remains to be evaluated.

Current evidence covers automated breadth and stateful fold-back on one retirement problem. Completing the machine requires a governed portfolio of target-blind world generation and target-aware oracle-premise decomposition, human-validated mechanism equivalence, concurrency-safe population control, diversity-preserving portfolio selection, and progressively realistic tests of surviving branches. The decisive next experiment is whether, across problems and models, world-level coverage control plus candidate-level ontology governance discovers useful mechanism classes missed by matched-compute direct, random-cue, random-world, and fixed-archive baselines. The larger wager is expansion before optimization: diversify the worlds in which a solution can become ordinary, measure what new mechanisms each world contributes, and preserve unusual survivors long enough to earn or lose evidentiary support.

Code and Artifact Availability

Code, paper source, prompts, graph databases, run artifacts, deterministic analysis scripts, and the trace-coding ledger are available at <https://github.com/emergent-wisdom/ontology-of-the-alien>. The separately released Orthogonal Insight Engine used for the world-diversity stage is available at <https://github.com/emergent-wisdom/orthogonal-insight-engine>; the earlier protocol paper is archived at doi:10.5281/zenodo.17905401. The original release of this paper is archived at doi:10.5281/zenodo.18912179.

References

- [1] Anil R. Doshi and Oliver P. Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28):eadn5290, 2024.
- [2] Jennifer Haase, Paul H. P. Hanel, and Sebastian Pokutta. Has the creativity of large-language models peaked? An analysis of inter- and intra-LLM variability. *Journal of Creativity*, 35(3):100113, 2025.
- [3] Yiming Zhang, Harshita Diddee, Susan Holm, Hanchen Liu, Xinyue Liu, Vinay Samuel, Barry Wang, and Daphne Ippolito. NoveltyBench: Evaluating language models for humanlike diversity. *arXiv preprint arXiv:2504.05228*, 2025.
- [4] Dustin Wright, Sarah Masud, Jared Moore, Srishti Yadav, Maria Antoniak, Peter Ebert Christensen, Chan Young Park, and Isabelle Augenstein. Epistemic diversity and knowledge collapse in large language models. *arXiv preprint arXiv:2510.04226*, 2025.
- [5] Mark A. Runco and Garrett J. Jaeger. The standard definition of creativity. *Creativity Research Journal*, 24(1):92–96, 2012.

- [6] Henrik Westerberg. Algorithmic creativity via strange worlds: A multi-agent toolkit for escaping the median trap. Zenodo, 2025.
- [7] Albert Einstein. Autobiographical notes. In Paul Arthur Schilpp, editor, *Albert Einstein: Philosopher-Scientist*, pages 1–95. Library of Living Philosophers, Evanston, IL, 1949.
- [8] Henrik Westerberg. Fractal intelligence: Conceptual decomposition as problem-solving infrastructure. Zenodo, 2026. Version 1; post-publication interpretive framework.
- [9] Edward de Bono. *Lateral Thinking: Creativity Step by Step*. Harper & Row, 1970.
- [10] Yining Lu, Dixuan Wang, Tianjian Li, Dongwei Jiang, Sanjeev Khudanpur, Meng Jiang, and Daniel Khashabi. Benchmarking language model creativity: A case study on code generation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2776–2794. Association for Computational Linguistics, 2025.
- [11] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 3982–3992. Association for Computational Linguistics, 2019.
- [12] Sentence Transformers. all-MiniLM-L6-v2 model card. Hugging Face, 2021. Model revision 1110a243fdf4706b3f48f1d95db1a4f5529b4d41.
- [13] John E. Arnold. Case study: Arcturus iv. MIT Creative Engineering Laboratory case study; Stanford Digital Repository, 1953. Original case study with later editorial commentary.
- [14] Brian Eno and Peter Schmidt. Oblique strategies: Over one hundred worthwhile dilemmas. Self-published card deck, 1975.
- [15] William J. J. Gordon. *Synectics: The Development of Creative Capacity*. Harper & Row, 1961.
- [16] Anthony Dunne and Fiona Raby. *Speculative Everything: Design, Fiction, and Social Dreaming*. MIT Press, 2013.
- [17] Dedre Gentner. Structure-mapping: A theoretical framework for analogy. *Cognitive Science*, 7(2):155–170, 1983.
- [18] Michihiro Yasunaga, Xinyun Chen, Yujia Li, Panupong Pasupat, Jure Leskovec, Percy Liang, Ed H. Chi, and Denny Zhou. Large language models as analogical reasoners. In *The Twelfth International Conference on Learning Representations*, 2024.
- [19] Pronita Mehrotra, Aishni Parab, and Sumit Gulwani. Enhancing creativity in large language models through associative thinking strategies. *arXiv preprint arXiv:2405.06715*, 2024.
- [20] Ruiying Ma, Chieh-Jan Mike Liang, Yanjie Gao, and Francis Y. Yan. MetaMuse: Algorithm generation via creative ideation. In *The Fourteenth International Conference on Learning Representations*, 2026. arXiv:2510.03851v1 submitted 4 October 2025.
- [21] Yuto Suzuki and Farnoush Banaei-Kashani. Universe of thoughts: Enabling creative reasoning with large language models. *arXiv preprint arXiv:2511.20471*, 2025. Version 1 submitted 25 November 2025.
- [22] Zhiwen Ruan, Yixia Li, Yefeng Liu, Yun Chen, Weihua Luo, Peng Li, Yang Liu, and Guanhua Chen. G2: Guided generation for enhanced output diversity in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 14116–14134. Association for Computational Linguistics, November 2025.
- [23] Rui Wang, Joel Lehman, Aditya Rawal, Jiale Zhi, Yulun Li, Jeffrey Clune, and Kenneth O. Stanley. Enhanced POET: Open-ended reinforcement learning through unbounded invention of learning challenges and their solutions. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 9940–9951. PMLR, 2020.
- [24] MiroFish contributors. MiroFish: Multi-agent social simulation repository. GitHub repository, 2025. Repository initiated in 2025; version inspected 10 August 2026.

- [25] Joel Lehman and Kenneth O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.
- [26] Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.
- [27] Herbie Bradley, Andrew Dai, Hannah Benita Teufel, Jenny Zhang, Koen Oostermeijer, Marco Bellagente, Jeff Clune, Kenneth O. Stanley, Grégory Schott, and Joel Lehman. Quality-diversity through AI feedback. In *The Twelfth International Conference on Learning Representations*, 2024.
- [28] Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, et al. Accelerating scientific discovery with co-scientist. *Nature*, 655:487–496, 2026. System publicly described in February 2025 as arXiv:2502.18864.
- [29] Runlin Duan, Nachiketh Karthik, Jingyu Shi, Rahul Jain, Maria C. Yang, and Karthik Ramani. ConceptVis: Generating and exploring design concepts for early-stage ideation using large language model. In *Volume 3B: 50th Design Automation Conference*, page V03BT03A042. American Society of Mechanical Engineers, 2024.
- [30] Pao Siangliulue, Joel Chan, Steven P. Dow, and Krzysztof Z. Gajos. IdeaHound: Improving large-scale collaborative ideation with crowd-powered real-time semantic modeling. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology*, pages 609–624. ACM, 2016.
- [31] Markus J. Buehler. Agentic deep graph reasoning yields self-organizing knowledge networks. *Journal of Materials Research*, 40:2204–2242, 2025.
- [32] Alireza Ghafarollahi and Markus J. Buehler. SciAgents: Automating scientific discovery through multi-agent intelligent graph reasoning. *arXiv preprint arXiv:2409.05556*, 2024.
- [33] Soheyl Massoudi and Mark Fuge. Agentic large language models for conceptual systems engineering and design. *arXiv preprint arXiv:2507.08619*, 2025.
- [34] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [35] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [36] Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, and Torsten Hoefler. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 17682–17690, 2024.
- [37] Michael T. Cox. Metacognition in computation: A selected research review. *Artificial Intelligence*, 169(2):104–141, 2005.

A Curated Trace Summaries

The following are author-curated summaries of saved model outputs, not verbatim transcripts and not validated proposals. The first three illustrate premise-level preconditioning and mechanism compilation; the fourth illustrates representation-level redirection by the candidate ontology.

Trace A: *theatrical* (F, run 10)

Seed and world	Witnessed acts acquire force; tension accumulates; public commitment matters more than accumulated quantity.
Alien solution	Retirement is a witnessed third act in which proportional sacrifice and narrative readiness determine claims.
Extraction	<i>Witnessed Sacrifice Retirement</i> : contributions are scored as a share of current variable income; hardship-adjusted contribution and witnessing replace a fixed monthly target.
Design move	Replace absolute amount with proportional effort.

Trace B: *desperacy* (F, run 22)

Seed and world	Authentic need attracts resources; synchronized needs multiply their pull, while satisfaction reverses and displaces need.
Alien solution	<i>Resonance Banking</i> : intergenerational groups preserve many mild needs that amplify an elder's urgent need; irregular earners' cycles keep this capacity active.
Extraction	<i>Resonance Pods: Variable-Bond Retirement</i> : small transparent groups make variable good-month contributions and coordinate money and care when an elder faces crisis.
Design move	Treat income volatility and hardship experience as coordination capacity rather than savings failure.

Trace C: *phosis* (F, run 24)

Seed and world	Luminous gradients force flow into a lower-radiance absorption well; different matrices preserve different kinds of value.
Alien solution	Capture is automatic. The Solver must instead design heterogeneous storage, a stabilizing anchor, reserve behavior, and integrity checks.
Extraction	<i>Absorption Well Retirement Collectives</i> : an at-source router contributes income above a moving personal baseline into separate financial, housing, health, and social-purpose pools.
Design move	Treat one bottleneck as provisionally solved, then derive the remaining architecture and seek a target-domain proxy for the stipulated capability.

Trace D: H18/H19 graph redirection

Occupied state	The graph already represented multi-pool distribution and personal-pattern scoring under <i>Distribute</i> and <i>Reframe Inconsistency</i> .
Rejection	A proposal combining overlapping pools with pattern-based priority was judged a mashup of those occupied mechanisms.
Diagnostic	The Taxonomist isolated one unrepresented relation: eliminate accumulation and the balance concept, retaining only claims on current flow.
Retry and restructure	<i>Balance-Free Flow Rights</i> allocates current inflow by duration credits; the admitted candidate created the <i>Dissolve Inconsistency</i> root.

Together, these traces illustrate two conditioning paths: a stipulated causal rule constrains the local solve before its operative relation is recast as an institutional mechanism, whereas an evolving ontology can reject an occupied combination and explicitly request another relation. The matched-source *limelike* F/H comparison appears in Section 4.3. Determining whether either path raises the finite-budget rate of original and useful mechanisms requires matched-compute evaluation.